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Introduction

Today, Russia plays a substantial role in shaping global energy markets and frequently dominates news headlines regarding natural gas exports and geopolitical actions. As one of the major suppliers of natural gas, Russia is in a unique position to leverage its energy resources as a tool of influence, especially in Europe. This has the potential to affect energy security, foreign policy, and electoral politics in many countries reliant on Russia for energy through the import of natural gas.

Through this paper, I seek to add to previous literature regarding the political and economic effects of Russian natural gas imports and voting by specifically focusing on vote shares for euroskeptic political parties in each country's national elections. While previous studies show that dependence on Russian natural gas can shape how European politicians vote in international institutions, less is known about whether that dependence also affects support for Euroskeptic parties at the national level. By combining Russian gas import data with party-level electoral data over time, this paper helps connect energy dependence to voter behavior and party competition in Europe. I intend to explore the following question: does dependence on Russian natural gas influence electoral support for Euroskeptic political parties in Europe?

To investigate this empirically, I assembled a panel dataset combining Russian natural gas imports from Eurostat with Chapel Hill Expert Survey (CHES) vote-share data for Euroskeptic parties in a set of European Union member states. I then estimated panel regressions with country and year fixed effects, controlling for log GDP per capita, natural gas import dependence, level of environmental concern, and major political events.

I will first give a brief overview on natural gas imports in the EU and Russia's natural gas industry. The second part of this paper will present the methods and empirical strategy and detail the regression results. The last parts will consist of discussion of results and a conclusion referencing potential broader implications of Russian energy dependence for electoral politics in Europe.

Literature Review

Thane Gustafson's *The Bridge* gives a detailed history of Russia's eventual natural gas dominance (2020). It began as a state-planned domestic venture back in the 60s and 70s. The USSR invested in natural gas following West Siberian gas discoveries in an attempt to counteract expected oil production declines and cultivate a new revenue stream. Europe's market for Russian gas developed through a series of "gas-for-pipe" deals where countries exchanged Soviet gas for Western technology and capital. This created a "gas bridge" that bound Europe to Russia through underground pipes and long term contracts meant to prioritize energy business over politics and ideology. Unlike traditional fossil fuels, natural gas is a "relationship commodity" since it transits through pipelines linking the producer and end consumer. This introduces many complex political factors into the gas market and also means that creating new deals, renegotiating, and switching sources is extremely challenging. While the relationship between Russia and the EU is currently tense, the "gas bridge" was originally intended as a business deal and not a geopolitical tool against the West, although it functioned as one against the Soviet satellites. This friction still exists with Ukraine, one of Russia's main transit partners, due to conflicts like the 2014 annexation of Crimea and the 2022 invasion of Ukraine.

Today, around 50% of all Russian natural gas passes through pipes under Ukraine on its way to international customers (Heuck, Kreysel, and Schiffer, 2025). That is a significant decrease from the 1990s, when Ukraine transited almost 90% of European imports of Russian natural gas. As European powers focus on environmental efforts to lower carbon emissions, one method they are using is replacing fuels like coal with natural gas as the "bridge fuel" to more environmentally friendly renewable energy (Perdana et al, 2022). By 2021, natural gas accounted for 24% of total European energy consumption, with 45% of that supplied by Russia (Heuck, Kreysel, and Schiffer, 2025). Following the 2022 invasion of Ukraine, EU member states aggressively diversified their import sources, bringing the Russian share down to 18.9% of total EU gas imports by 2024, largely through a rapid expansion of liquified natural gas imports from the United States.

Korteweg details the use of energy for foreign policy interests by authoritarian states like Russia. He argues that natural gas is used as a strategic foreign-policy lever to reward allies and punish opponents in Europe (2018). Russia has historically used targeted pricing to provide discounts to allies and higher prices to opponents seeking to receive imports. Additionally,

Russia also uses direct threats of disruptions and cutoffs to increase the political vulnerability of transit partner states and destination states lacking in energy source diversification. Furthermore, the long term contracts detailed by Gustafson can also be used to limit flexibility and ability of EU countries to diversify their energy options or renegotiate. Korteweg's paper suggests that flows of Russian gas through Europe are shaped by politics and have the ability to serve as a tool of geopolitical influence, reinforcing Russia's bargaining power and shaping the foreign-policy choices of dependent European states. Using a panel of European countries from 1995 to 2013, they find that higher natural gas dependence on Russia correlates with more pro-Russian voting behavior in the United Nations, suggesting that energy dependence shapes the political behavior of importing states. Their findings establish a foreign policy channel through which dependence operates; this paper extends that logic to the domestic electoral arena.

Given the foreign policy impacts and disruptions to citizens in destination countries, Russia's manipulation of cost and flow of natural gas has the potential to impact electoral decisions of individuals and government actors. Chadwick and Long build on Korteweg's finding that, even in the absence of clear coercion, Russian natural gas raises the cost of exit to such a level that importing countries become less willing to criticize Russia or adopt oppositional political positions. They explore this relationship through empirical panel data regressions of UN general assembly voting patterns in 27 European countries from 1995 to 2013 (2023). Their results show that greater dependence on Russian natural gas is associated with more similar voting behavior toward Russia at the UNGA (Chadwick & Long 2023).

Moraski and Giurcanu make a similar argument through their study of Russian gas dependence and its effect on politicians' responses to Russia's 2008 invasion of Georgia, using voting data from the Parliamentary Assembly of the Council of Europe (PACE) (2013). In their study, they found that European politicians from countries more dependent on Russian natural gas were less likely to support resolutions condemning Russia's actions, while countries with little dependence sometimes took a harsher stance. These results suggest that Russian energy dependence can shape political decisions by discouraging criticism, but the relationship is not entirely straightforward (Moraski & Giurcanu 2013). They also found that some highly dependent countries still chose to condemn Russia, which suggests that additional political and regional factors also play a role.

In regards to euroskeptic voting patterns, one major contributor could be the rise in support for populist parties and Europe in the past 10 to 15 years. This has been fueled by several social and economic issues ranging from financial crises, inequality, and increased immigration (Fischer and Meister 2023). Citizens of EU countries have become disillusioned with leadership and have instead turned to parties like Brothers of Italy in Italy, Jobbik in Hungary, PiS in Poland, and the AfD in Germany. Right-wing populist parties are able to appeal to people's anger and distrust of institutions and current "elites" and provide an alternative. As a result, these parties position themselves against current EU infrastructure and take a euroskeptic stance. This has increased euroskeptic voting in states with larger vote share for populist far-right parties.

Data Description

The complete, merged annual country-year panel dataset consists of all current EU member states: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden. Since the United Kingdom left the EU and does not have consistent data for all variables, it has been removed.

The dataset is constructed by combining multiple variables from different sources to create a panel of interest and control variables spanning three decades. The two main variables are percent of natural gas imported from Russia in gaseous form and euroskeptic voteshare. Gas import ratios are found by calculating the proportion of gaseous and liquid natural gas a country uses sourced from Russia out of their total amount of imported natural gas. Russian gas imports are reported in both gas form that travels through pipes and liquid form that is stored in vats and can be transported a number of ways. All gas import data is sourced from Eurostat and is reported in millions of cubic meters.

Natural gas transported through pipelines has more of a political impact because of long term contracts and difficulty switching providers. Additionally, data on liquid natural gas is a lot more sparse because of how complicated its supply chain can be and because transferring the gas form through underground pipes is the older, more established method. Because of this, gaseous natural gas ratio is explored more in depth. I will explore the liquid form through alternative

metrics in the discussion, since the literature on energy sourcing and transport frequently highlights liquefied natural gas as a potential solution to dependence on Russian supplies.

Euroskeptic voteshare percent is calculated using values from the Chapel Hill Expert Survey. I created the variable “euroskeptic_share” to represent the overall vote share for euroskeptic parties for each country-year combination. It is based on the CHES EU_POSITION variable, where parties scoring 3 or less were coded as euroskeptic and given a binary value of 1, while all others were coded as 0. The final euroskep measure is then calculated by adding the vote share of all parties classified as euroskeptic. Since this variable is only measurable in years with national elections, data for in-between years will be forward filled with the last available euroskeptic voteshare.

The remaining variables account for additional factors that could influence the relationship between imported Russian gas and voting. The variables “imp_dep_ngg” and “imp_dep_lng” measure each country’s overall natural gas import dependency in gaseous and liquid form as the share of energy needs met by foreign imports, with higher values meaning greater dependence and negative values meaning net exports.

Environmental priority of governments is represented by “envs_share” which measures the vote share of parties that prioritize environmental protection over economic growth, based on CHES environment scores of 4 or lower.

GDP per capita is sourced from Our World In Data reported in constant 2021 USD. Event variables, “post2008”, “post2014”, and “post2022” indicate if a year is after the Russian invasions of Georgia, Crimea, and Ukraine, respectively. The variable “eu_debt” represents a period of years that could have been impacted by the European debt crisis. “eu_debt” codes country-year pairs in 2010 through 2013 with a 1. This range was chosen because the earlier years were more likely to be affected by the initial shock and future years were more likely to have less of an effect and potentially be influenced by other events. 2014 was purposefully not included in “eu_debt” because it is accounted for in post2014 and there could be overlaps of the debt crisis and the invasion of Crimea.

Countries that are closer to Russia tend to receive more natural gas imports. This makes sense due to the decreased transit time and distance. The distances between countries are controlled through the use of country fixed effects.

Methodology

To explore whether dependence on Russian gas affects support for parties skeptical of the European Union, I estimated panel regression models using country and year fixed effects. The basic model is as follows:

$$\begin{aligned} \text{EuroskepticShare}_{it} = & \beta_0 + \beta_1 \text{EuroskepticShare}_{i,t-1} + \beta_2 \text{NGGRatio}_{i,t-1} + \\ & \beta_3 \text{NggImpDep}_{it} + \beta_4 \text{LogGDPpc}_{it} + \delta_t + \alpha_i + \epsilon_{it} \end{aligned}$$

$\text{EuroskepticShare}_{i,t-1}$ is the lagged voteshare of Euroskeptic parties in country i and year t . It accounts for autocorrelation between the voteshare each year and the previous year. NGGRatio_{it} measures dependence on Russian natural gas. Models experiment with using NGGRatio_{it} of the current year as well as the lagged form to account for the gas data being representative of a whole year and elections occurring in different months and parts of the year. NggImpDep_{it} represents the how dependent country i is on imported gaseous natural gas in year t .

I then add a series of control variables to build a thorough model account for potential omitted variable bias. LogGDPpc_{it} is the log GDP per capita of country i in year t . Event variables “eu_debt”, “post2008”, “post2014”, “post2022” are included in later, more complex models to control for the European Debt Crisis and major political events of the 2008 invasion of Georgia, the 2014 annexation of Crimea, and the 2022 invasion of Ukraine. These variables target suspected shocks and important events.

To see if being a member of the Eurozone had any effect on euroskeptic voteshare, I added a dummy variable called “euro” to indicate if a country adopted and used the euro at any time within the data time frame. This is because countries who share a common EU currency may be less euroskeptic. Country fixed effects α_i account for time-invariant differences across countries and year fixed effects δ_t account for shocks that affect all countries in a given year. ϵ_{it} is the error term.

The main assumption is that the model accounts for major variables responsible for modeling the relationship between Russian gas use and dependence and Euroskeptic voteshare. The results of the regression are intended to show correlation, but not necessarily causality.

Results

Figure 1

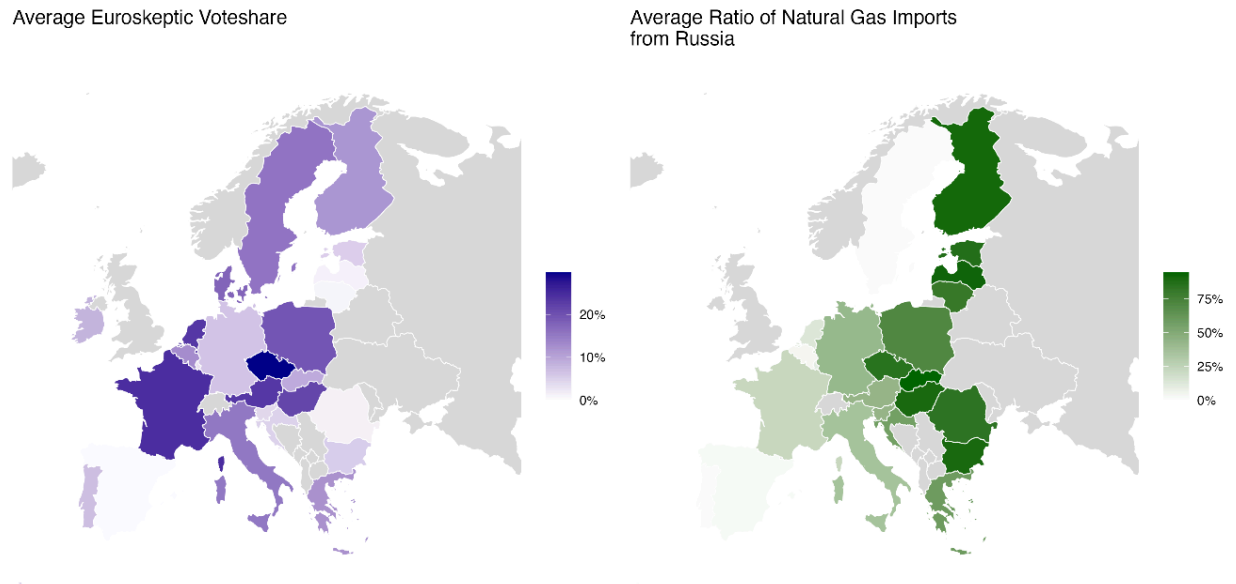


Figure 1 shows the average euroskeptic voteshare of each country next to the average ratio of gaseous natural gas imported from Russia of each country. Euroskeptic voteshare appears to be more geographically varied with countries like France and Hungary being more euroskeptic and countries like Romania, Latvia, and Lithuania being less euroskeptic. The average ratio of natural gas imported from Russia appears to have a strong geographical correlation with how far a country is from Russia. Countries that border the former USSR states of Ukraine, Russia, and Belarus, have much larger average ratios. This pattern appears to be layered with countries like Germany and Italy having a moderate ratio of Russian gas imported and the Netherlands, Belgium, and France importing a much smaller quantity on average.

Figure 2

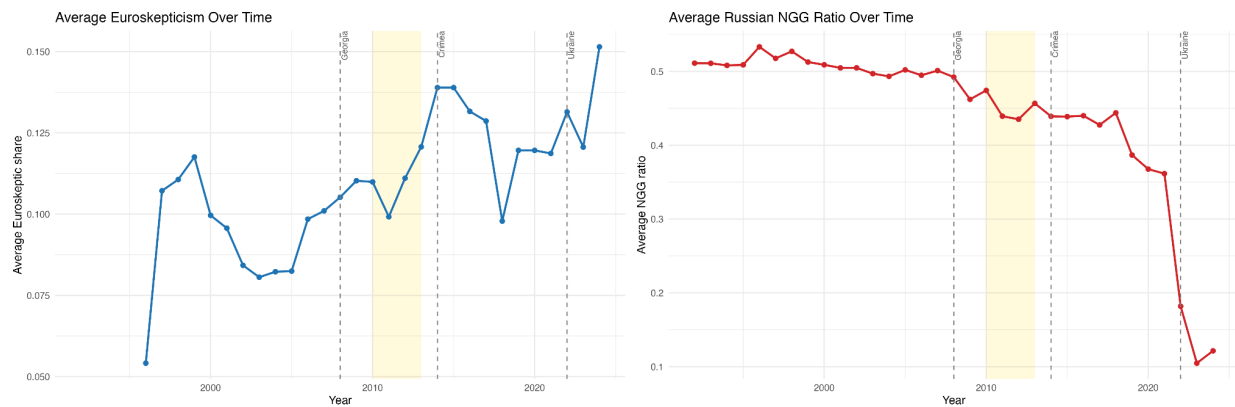


Figure 2 plots the average euroskeptic voteshare for all countries next to the average ratio of imported gaseous Russian natural gas for all countries over time. Dashed lines represent potential important political events and the yellow shaded portion represents the years of 2010 through 2013 where there could be some effect from the European debt crisis. There appear to be spikes in euroskepticism around 1998, 2014, and 2022, with 2014 and 2022 correlating with Russian military action in Ukraine. Since the European debt crisis occurred over several years, 2014 also overlaps with economic timelines. There appear to be dips in euroskepticism in 2002, 2018, and 2023. Looking at the average ratio of imported gaseous natural gas from Russia, there appears to be an increasing downward trend with a noticeable decrease during the highlighted portion representing the European debt crisis.

Table 1

Regressors	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
L.euroskeptic_share			0.745***	0.745***	0.742***	0.745***	0.742***	0.732***
			-(0.0296)	-(0.0296)	-(0.0298)	-(0.0296)	-(0.0296)	-(0.031)
L.ngg_ratio			-0.0269*	-0.0272*	-0.0249*	-0.0270*	-0.0283**	-0.0205
			-(0.0143)	-(0.0146)	-(0.0145)	-(0.0144)	-(0.0144)	-(0.0158)
imp_dep_ngg		0.000401**	0.000142	0.000142	0.000112	0.000143	0.000141	7.32E-05
		-(0.000197)	-(0.000145)	-(0.000145)	-(0.000149)	-(0.000145)	-(0.000148)	-(0.000157)
log_gdppc		0.0353	0.00878	0.00956	0.00499	0.0087	0.00744	-0.025
		-(0.0231)	-(0.0173)	-(0.0186)	-(0.0177)	-(0.0174)	-(0.0178)	-(0.025)
ngg_ratio	-0.0456***	-0.0323*						
	-(0.0168)	-(0.0186)						
euro				-0.0013				
				(-0.0113)				
envs_share					0.0183			
					(-0.0193)			
imp_dep_lng						-5.27E-06		
						(-0.000143)		
eu_debt							0.00477	
							(-0.00678)	
post2008							0.000723	
							(-0.0128)	
post2014							0.0243**	
							(-0.0119)	
post2022							0.0113	
							(-0.0124)	
Constant	0.131***	-0.281	-0.062	-0.0694	-0.0217	-0.061	-0.0489	0.29
	-(0.00775)	-(0.248)	-(0.185)	-(0.197)	-(0.19)	-(0.188)	-(0.191)	-(0.268)
Observations	644	644	617	617	617	617	617	617
R-squared	0.012	0.024	0.531	0.531	0.531	0.531	0.535	0.552
Number of country_id	27	27	27	27	27	27	27	27
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time Fixed Effects	No	No	No	No	No	No	No	Yes
Event Dummies	No	No	No	No	No	No	Yes	No

In the naive model, column (1), the relationship between the ratio of natural gas imported from Russia and the euroskeptic voteshare is significant at the 1 percent level with a negative coefficient. This suggests that an increase in gaseous natural gas imports is correlated with a decrease in euroskeptic voteshare. Although this is very significant in the first regression, the R-squared value is very small, meaning the model does not explain much about the relationship between imports and voteshare. The significance of this finding also wanes as more variables and controls are added to later models.

In model (2), dependence on imported gaseous natural gas is added to the model. The variable has a very small positive coefficient that is significant at the 5 percent level suggesting that higher Russian gas dependence is associated with slightly higher euroskeptic vote share.

For models (3) through (8), lagged euroskeptic vote share is added and very significant at the 1 percent level. The R-square values are much higher, indicating that the model is a much better fit compared to the naive regression. At face value, this suggests that there is a high level of autocorrelation between years, but since elections are only every 4 years, forward filling the data to build the panel set could have contributed to this finding. As a result, there could definitely be correlation between years, but it is impossible to get a conclusive answer due to the format of the data.

I choose to experiment with lagged variables since gas data is reported as a total for each year and election cycles can happen in different months. Lagged gaseous natural gas imports are slightly significant at the 10 percent level in models (4) through (6) and significant at the 5 percent level in model (7). The small, negative coefficient indicates that a rise in dependence on gaseous Russian natural gas imports from the previous year is correlated with a decrease in the next year's euroskeptic voteshare. This is in line with findings for non-lagged gaseous natural gas imports.

Controls for being in the Eurozone, how much a government prioritizes environmental concerns compared to economic ones, and import dependency on specifically liquified natural gas were added one by one in models (4), (5), and (6). While using the Euro could play a role in impacting euroskeptic voting, the variable was not found to be significant. Since some countries are using natural gas as a bridge fuel from coal to clean energy, there is potential for people's votes for environmentally conscious candidates to impact their views on energy, natural gas, wider EU politics, and regarding euroskepticism. In this case, the vote share for environmental leaning parties was not significant in the model. Furthermore, while the dependence on gaseous natural gas was significant in model (2), the dependence of a country on liquified natural gas was not significant. While all three of these controls were found not to be significant, they act as robustness checks to make sure that other variables that were mentioned in previous literature and could influence euroskeptic voting were not left out.

The last two models, (7) and (8) were used to see if certain years or specific events changed results. In model (7), country fixed effects were used along with the following dummy

variables: “eu_debt”, “post2008”, “post2014”, and “post2022”. The EU debt crisis was added to see if financial stress and economic turmoil changed people’s perception of the EU and either increased or decreased euroskeptic voting. Of these dummies, only “post2014” was found to be significant at the 5 percent level. This means that there is a difference in euroskeptic voting after 2014 and euroskeptic voting is a bit higher due to the small positive coefficient. This is interesting to note, since it indicates there could be a change, event, or shift that is impacting the data, but the specifics are hard to tease out due to the overlap of the debt crisis and the invasion of Crimea in 2014.

Model (8) does not use dummy variables for targeted events and shocks and instead just pairs country fixed effects with time fixed effects. None of the individual years were found to be significant and in this regression, the only variable of note is lagged euroskeptic share which is significant at the 1 percent level.

Discussion

While the dataset and models contain many control variables, one limitation of the study was I was unable to account for countries who were recipients of the EU cohesion fund. The Cohesion Fund provides financial support to member states who have a gross national income per person that is under 90% of the average total of all 27 EU member states with the goal of strengthening the overall unity of the EU from social, economic, and territorial standpoints (European Commission). The EU provides a publicly accessible database on Cohesion Fund allocation, but aggregates the data in many different ways. Furthermore, the data only covers 2014 to 2027, the second half of my year range. As a result, I was unable to create a variable to control for Cohesion Fund recipients. This has the potential to cause some omitted variable bias since countries who are receiving funding from the EU might have lower euroskeptic voteshares.

Another potential source of bias was through my limited exploration of environmental factors relating to Euroskeptic voting. While I was able to include a variable in one of my robustness checks to measure the impact of each country’s environmental voteshare, my research would have been more complete if I was able to get yearly data on each country’s environmental goals and policies. Recently, there has been a lot of focus on the transition to green energy in the EU. One strategy is using natural gas as a temporary “bridge fuel” instead of coal. This could cause bias in my research since countries could be prioritizing green energy and altering their

natural gas usage as a result. Voters who wish to prioritize the government might favor EU wide environmental policies and be less euroskeptic. On the other hand, citizens who do not support the energy transition could become more euroskeptic in an attempt to distance themselves from EU peers and wider EU governing coalitions who wish to undertake a joint green energy venture.

While I was able to access data on liquified natural gas imports, the data was a lot more sparse and limited compared to the gas form. For the main research question, the impact of gaseous natural gas is a lot more impactful to euroskeptic voting since a majority of it is sourced from Russia through pipelines and contracts that are not easily changed. Because of that, this paper focused mostly on gas imported through pipes. It is important to acknowledge, however, that the increasing ease of transporting and trading liquified natural gas has the potential to alter the complex political and economic relationship that ties Russia to the EU through the “gas bridge”. In the future, further research specifically regarding the impact of access to liquified natural gas to allow for diversification of providers could be useful to include as a control variable. Additionally, I was unable to get reliable data on storage of natural gas in either form. This is a potential source of bias since countries could import less gas if they already have a stockpile.

Conclusion

In conclusion, there appears to be a correlation between Russian gaseous natural gas imports to EU countries and euroskeptic voting. The negative coefficients on the lagged and normal measures of gas import ratio suggest an increase in natural gas imports is correlated with a decrease in euroskeptic voteshare. This is counterintuitive to previous literature measuring UNGA votes and could mean that while Russian natural gas imports may cause politicians and government officials to vote more euroskeptically, this does not translate to the general populace. Additionally, the significance of the “post2014” dummy suggests there could have been an important event that caused an important shift to differentiate years leading up to 2014 and the years after. While lagged euroskeptic share is significant and indicates autocorrelation between years, this finding is muddled by the format of the data where years without national elections were forward filled with data from the most recent election. Overall, this paper suggests that Russian gas imports are politically relevant in EU country national elections but the effect on euroskeptic voting is somewhat mixed and sensitive to model specification.

Bibliography

European Commission. 2024. "Cohesion Fund." Accessed May 13, 2026.

https://ec.europa.eu/regional_policy/funding/cohesion-fund_en.

Eurostat. 2024. "Energy Statistics — An Overview." European Commission (Eurostat), May 21, 2024.

https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Energy_statistics_-_an_overview.

Eurostat. 2026. Energy Imports Dependency: Custom Table nrg_ind_id__custom_20841109.

Luxembourg: Eurostat.

https://ec.europa.eu/eurostat/databrowser/view/nrg_ind_id__custom_20841109/default/table.

Fischer, Carl Leonard, and Lorenz Meister. 2023. "Economic Determinants of Populism." DIW Roundup 145. Berlin: DIW Berlin.

https://www.diw.de/de/diw_01.c.879785.de/publikationen/roundup/2023_0145/economic_determinants_of_populism.html.

Gustafson, Thane. 2020. *The Bridge: Natural Gas in a Redivided Europe*. Cambridge, MA: Harvard University Press.

Stoelzel Chadwick, Christina M, and Andrew G Long. 2023. "Foreign Policy Alignment and Russia's Energy Weapon." *Foreign Policy Analysis* (Oxford) 19 (2): 1.

<https://doi.org/10.1093/fpa/orac042>

Heuck, Lukas, Mark-Alexander Kreysel, and Hans-Wilhelm Schiffer. 2025. "Geopolitics and Europe's Natural Gas Supply: Aspects and Consequences from a Gas and Electricity Markets Perspective." *Inter Economics* (Warsaw) 60 (6): 346–54.

<https://doi.org/10.2478/ie-2025-0068>.

Korteweg, Rem. 2018. *Energy as a Tool of Foreign Policy of Authoritarian States, in Particular Russia*. European Parliament. <https://doi.org/10.2861/951739>.

Moraski, Brian, and Mihaela M. Giurcanu. 2013. "Elections, Empowerment, and Legislative Voting in the Parliamentary Assembly of the Council of Europe (PACE)." *Journal of Legislative Studies* 19 (3): 379–403. <https://doi.org/10.1080/13572334.2013.785635>.

Perdana, Sigit, Marc Vielle, and Maxime Schenckery. 2022. "European Economic Impacts of

- Cutting Energy Imports from Russia: A Computable General Equilibrium Analysis.”
Energy Strategy Reviews (St. Louis) 44 (November): 101006.
<https://doi.org/10.1016/j.esr.2022.101006>.
- Rovny, Jan, Ryan Bakker, Liesbet Hooghe, Seth Jolly, Gary Marks, Jonathan Polk, Jan Rovny, Marco Steenbergen, and Milada Anna Vachudova. 2025. "The 2024 Chapel Hill Expert Survey on Political Party Positioning in Europe: Twenty-Five Years of Party Positional Data." Electoral Studies 97. <https://doi.org/10.1016/j.electstud.2025.102981>. Data: CHES Trend File 1999–2024, Version 2025.21. <https://chesdata.eu>.
- Roser, Max, and Esteban Ortiz-Ospina. 2024. "GDP per Capita." Our World in Data. <https://ourworldindata.org/grapher/gdp-per-capita-worldbank>.
- U.S. Energy Information Administration. 2026. “Europe Relies Primarily on Imports to Meet Its Natural Gas Needs.” Today in Energy, March 9, 2026.

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